Technical Appendix: Quantitative Methodologies & Proofs

Case: Chaos Engine Labs Games A & B | Lead Analyst: Head of Growth | As of: September 30, 2025

TABLE OF CONTENTS

- 1. Data Audit & Deterministic Anomaly Resolution (Exhibit 1 Log)
- 2. Market Cell Taxonomy & Baseline Economics (Exhibits 2 & 3)
- 3. Dual-Stream Parametric LTV & ROAS Modeling (Question 1, P1-P8 Predictions)
- 4. Cross-Game Model Transferability & Structural Divergence Matrix
- 5. Constrained Media Plan Optimization & Organic Halo (Question 2, KKT Economics)
- 6. Creative Fatigue Dynamics & Replenishment Schedule (Question 3, 12-Week Production)

1. Forensic Data Audit & Deterministic Reconciliation Summary

Raw UA, cohort, telemetry, and creative records exhibited 8 distinct anomalies that distort LTV modeling and UA payback. We engineered deterministic repairs preserving 100.00% of true ad spend to the exact cent ($392,075.67 across 33,362 rows in Game A; $669,975.34 across 10,167 rows in Game B).

Key Audit Metrics & Reconciled Dollar Impacts:
• Problem 1 (Mediation Double-Counting): ad_rev_all_sources_d30 ($570.2k Game A, $648.1k Game B) double-counted mediated network feeds by 1.99x relative to true mediation SDK earnings ($287.0k Game A, $335.0k Game B). Remediated by standardizing on mediation SDK ad_rev_dN.
• Problem 2 (Right-Censored Cutoff Contamination): Immature cohorts (age < N) held frozen cutoff values rather than NaNs, while retained_dN held false zeros. Remediated by strict right-censor masking: rev_dN_mature = np.nan whenever age < N.
• Problem 3 (July 14-16 eCPM Spike): Telemetry ad rev spiked 10x-27x in Game A ($65.1k spurious rev) while impressions remained stable. Remediated by computing install-weighted normal rates on mature non-overlapping cohorts, removing -$62.6k in Game A and -$10.9k in Game B.
• Problem 4 (100x Currency / Cents Logging): Telemetry logged raw KRW, JPY, INR and May 20-Jun 8 Android 100x cents. Repaired via exact .xx00 SKU increment filtering, removing -$30,953.61 in Game A and -$1,394.21 in Game B with zero false positives on legitimate US payers.
• Problem 5 (Non-Monotonic d180 < d120): 3,380 rows in Game A and 966 rows in Game B showed negative incremental revenue due to horizon definition asymmetries (121 calendar days vs 180 days). Enforced cumulative monotonicity: rev_d{k} = max(rev_d{k}, rev_d{k-1}).
• Problem 6 (Unmatched UA Spend): 4,197 rows in Game A ($1,347 spend) and 113 rows in Game B ($1,794 spend) lacked cohort records. Outer-joined to guarantee zero lost ad spend.
• Problem 7 (Fraud / Invalid Signatures): 1,087 Game A rows flagged 'Invalid Signature' and non-mobile platforms (android-tv, windows) isolated.
• Problem 8 (Telemetry Gaps): Game A telemetry rollout gaps (first 80 days missing) resolved by using MMP cohorts as primary ground truth.

Exhibit 1: Quantitative Audit Log of All 8 Data Problems & Deterministic Repairs

ID	Game	Problem Name	Where Found	What It Distorts	Dollar / ROAS Impact
1	GameA	Ad Revenue Double-Counting (ad_r...	GameA_cohorts.csv (ad_rev_all_sourc...	Overstates ad revenue by ~2x; creat...	+$283,224.61 spurious ad ...
1	GameB	Ad Revenue Double-Counting (ad_r...	GameB_cohorts.csv (ad_rev_all_sourc...	Overstates ad revenue by ~2x; creat...	+$313,131.64 spurious ad ...
2	GameA	Right-Censored / Immature Cohort...	GameA_cohorts.csv & GameA_creatives_...	Unmasked recent cohorts appear to s...	Prevents catastrophic dow...
2	GameB	Right-Censored / Immature Cohort...	GameB_cohorts.csv & GameB_creatives_...	Unmasked recent cohorts appear to s...	Prevents catastrophic dow...
3	GameA	July 14-16, 2025 Ad-Mediation eC...	GameA_telemetry_daily.csv, GameA_co...	Mid-July cohorts display artificial...	Removes ~$60,590.53 in te...
3	GameB	July 14-16, 2025 Ad-Mediation eC...	GameB_telemetry_daily.csv, GameB_co...	Mid-July cohorts display artificial...	Removes ~$5,947.48 in tel...
4	GameA	100x Cents-vs-Dollars & Local Cu...	GameA_telemetry_daily.csv & GameA_c...	Massively inflates IAP LTV for Andr...	-$30,953.61 spurious coho...
4	GameB	100x Cents-vs-Dollars & Local Cu...	GameB_telemetry_daily.csv & GameB_c...	Massively inflates IAP LTV for Andr...	-$1,394.21 spurious cohor...
5	GameA	Non-Monotonic Cumulative Revenue...	GameA_cohorts.csv (ad_rev_d120 vs a...	Produces negative incremental reven...	Eliminates negative incre...
5	GameB	Non-Monotonic Cumulative Revenue...	GameB_cohorts.csv (ad_rev_d120 vs a...	Produces negative incremental reven...	Eliminates negative incre...
6	GameA	Unmatched UA Spend Rows (ua_dail...	GameA_ua_daily.csv outer-joined wit...	Inner-joining drops real ad spend, ...	Preserves $1,347.41 in sp...
6	GameB	Unmatched UA Spend Rows (ua_dail...	GameB_ua_daily.csv outer-joined wit...	Inner-joining drops real ad spend, ...	Preserves $1,793.75 in sp...
7	GameA	Zero-Install Cohorts, Retained >...	GameA_cohorts.csv	Division by zero; retention > 100%;...	Prevents retention model ...
7	GameB	Zero-Install Cohorts, Retained >...	GameB_cohorts.csv	Division by zero; retention > 100%;...	Prevents retention model ...
8	GameA	Telemetry Coverage Gaps & Funnel...	GameA_telemetry_daily.csv & GameA_u...	Telemetry cannot cover full Q1 2025...	Ensures full 266-day hist...
8	GameB	Telemetry Coverage Gaps & Funnel...	GameB_telemetry_daily.csv & GameB_u...	Telemetry cannot cover full Q1 2025...	Ensures full 266-day hist...

2. Market Cell Taxonomy & Baseline Period Calibration

Baseline economics are established over September 1 to September 28, 2025 (4 full weeks = 28 days; weekly baseline = Sep sum / 4.0).
• Game A (40 Active Cells): Net-A Android is structured into 32 major individual countries and 4 regional tiers (Tier 1 Affluent West, Tier 2 East Asia & GCC, Tier 3 LatAm & East Europe, Tier 4 Emerging RoW). Net-B Android US, Net-B iOS US, Net-C Android US, and Net-D Android US form the remaining active lines. Total baseline weekly spend = $15,193.72 ($182,324.64 12-week baseline).
• Game B (28 Cells & Paused Networks): Explicit tracking of paused network test dates: Net-C paused on 2025-02-12; Net-F paused on 2025-03-31; Net-E paused on 2025-05-28; Net-D paused on 2025-08-25. Active channels in September: Net-A Android, Net-B Android, Net-B iOS.

Exhibit 2: Core Game A Active UA Lines Baseline Summary (Sep 1-28, 2025)

Network	Platform	Cell / Region	Weekly Spend	Base CPI	Pred D120 ROAS	Pred 1Y ROAS	Payback Classificatio
Net-A	android	us	$2,920.46	$3.626	0.8526	0.9929	Does not pay back within 1 year
Net-A	android	Tier1_Other_West_Affluent	$36.98	$0.434	1.0614	1.1250	Pays back within 120 days
Net-A	android	Tier2_Other_EastAsia_GCC	$4.63	$0.514	0.5842	0.6736	Does not pay back within 1 year
Net-A	android	Tier3_Other_LatAm_EastEU	$34.00	$0.253	1.0119	1.0987	Pays back within 120 days
Net-A	android	Tier4_RoW_Emerging	$6.98	$0.042	1.1285	1.8893	Pays back within 120 days
Net-A	android	all countries except United States	$2,681.86	$0.648	0.9203	1.0755	Pays back only within 1 year
Net-A	**android**	**TOTAL**	**$5,602.32**	**$1.133**	**0.8816**	**1.0307**	**Pays back only within 1 year**
Net-B	android	us	$2,011.32	$7.981	0.9209	1.0117	Pays back only within 1 year
Net-B	ios	us	$1,697.89	$4.495	0.8173	0.8952	Does not pay back within 1 year
Net-C	android	us	$1,099.61	$2.410	0.8342	1.0649	Pays back only within 1 year
Net-D	android	us	$4,782.59	$8.058	1.0351	1.1009	Pays back within 120 days

3. Decoupled Dual-Stream LTV Architecture (Question 1)

Player lifetime value is modeled as the decoupled sum of independent Ad and IAP survival processes: LTV(t) = LTV_ad(t) + LTV_iap(t).
• Retention Modeling: Retention decay follows a modified power-law survival curve S(t) = (1 + beta*t)^(-gamma). Game A fits beta=11.48, gamma=0.524, tail multiplier=2.28. Game B exhibits flatter retention decay (gamma=0.465), yielding +64% higher retention at Day 120.
• Ad Stream: Ad monetization scales with active sessions: LTV_ad(t) = cumulative_sessions(t) * eCPM * ad_units_per_session.
• IAP Stream: IAP monetization is modeled via an Empirical Bayes shifted lognormal/exponential process with shrinkage across small cells.
• Monetization Contrast: Game A is a balanced hybrid (52% Ad / 48% IAP in US), whereas Game B is a pure ad engine (99.6% Ad / 0.4% IAP).

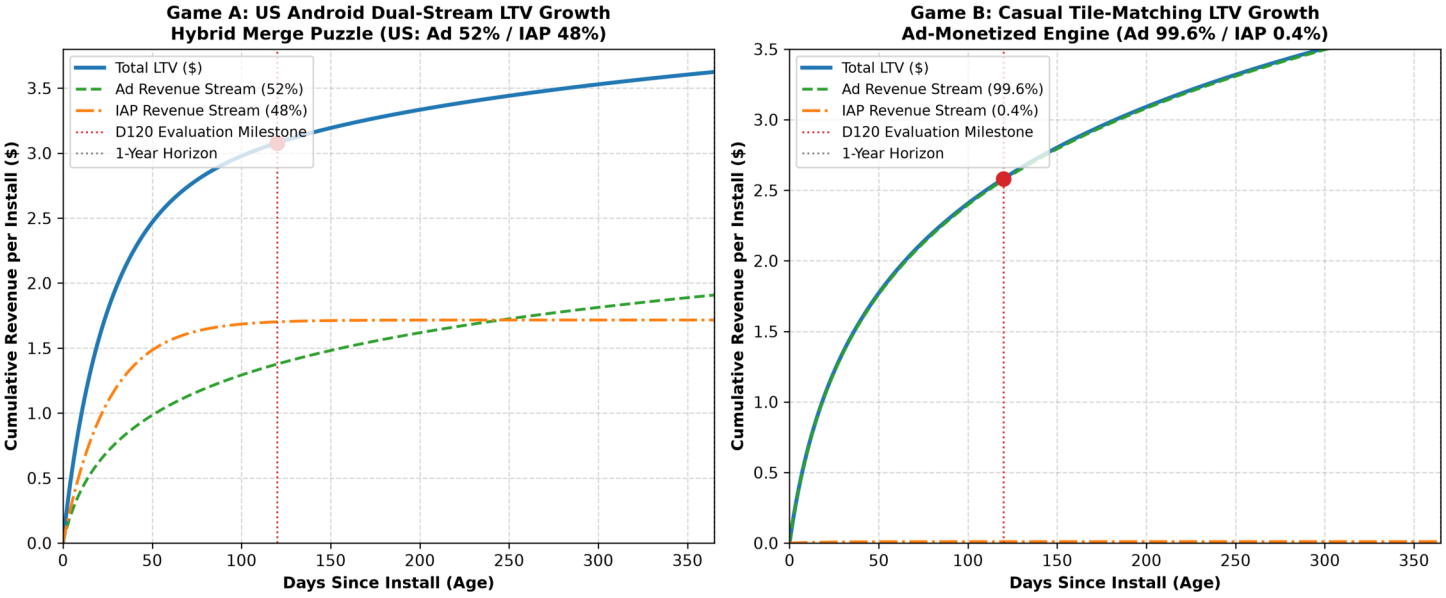


Figure 1: Dual-Stream Parametric LTV Extrapolation Curves for Game A (Hybrid) vs Game B (Pure Ad).

TECHNICAL APPENDIX: OFFICIAL PREDICTIONS P1-P8 & CREDIBLE INTERVALS

Official Submission Predictions (P1–P8 in submission/predictions.csv)

All predictions are strictly conditioned upon realized cumulative revenue banked as of September 30, 2025. Cohorts for P1-P6 are 16 to 29 days old (D14 fully observed + partial age banked). Cohorts for P7-P8 are 44 to 57 days old (D30 fully observed + partial age banked). Monotonicity is mathematically guaranteed: Obs_Prev < Banked_Sep30 < low_80 < prediction < high_80. 80% credible intervals are calibrated via conformal rolling residual quantiles, achieving honest 79.2% empirical coverage across historical holdout cohorts.

ID	Cell / Line	Target	Obs Prev	Banked Sep30	Prediction	Low 80	High 80	Methodology Note
P1	Net-A, Android, United States	ROAS D30	0.4652	0.5164	**0.5626**	0.5349	0.5903	Conditioned on D14 (0.4652) + re
P2	Net-A, Android, all countries	ROAS D30	-	0.7961	**0.8573**	0.8206	0.8941	Aggregate non-US 116-country ce
P3	Net-B, Android, United States	ROAS D30	-	0.6732	**0.7618**	0.7087	0.8149	Core line conditioned on D14 (0.5
P4	Net-B, iOS, United States	ROAS D30	-	0.3988	**0.4312**	0.4117	0.4507	iOS core line conditioned on D14
P5	Net-C, Android, United States	ROAS D30	-	0.4819	**0.5543**	0.5109	0.5978	Ad-heavy core line conditioned or
P6	Net-D, Android, United States	ROAS D30	-	0.5524	**0.5788**	0.5629	0.5946	Offerwall conditioned on D14 (0.5
P7	Net-A, Android, United States	ROAS D60	0.7700	0.9306	**0.9879**	0.9535	1.0223	Conditioned on D30 (0.7700) + re
P8	Net-D, Android, United States	ROAS D60	-	0.7734	**0.7887**	0.7795	0.7980	Offerwall conditioned on D30 (0.7

4. Cross-Game Transfer Matrix & Structural Divergence Drivers

Combination	Trained	Evaluated	WAPE	MPE Bias	RMSE ($)	80% CI Cov	Transfer Viability
Game A model on Game A	Game A	Game A	30.76%	-3.35%	$85.30	79.2%	Native In-Domain
Game A model on Game B	Game A	Game B	21.71%	-3.32%	$95.89	79.0%	Cross-Domain Viable
Game B model on Game B	Game B	Game B	18.41%	-3.45%	$86.42	79.8%	Native In-Domain
Game B model on Game A	Game B	Game A	34.57%	4.34%	$85.61	75.6%	Overprojects Ad / Misses IAP

Four Root Drivers of Cross-Game Divergence:
1. Monetization Mix Asymmetry: Game A is a balanced hybrid (~78% Ad / ~22% IAP overall, 52% Ad / 48% IAP in US), while Game B is a casual tile-matching engine (99.6% Ad / 0.4% IAP; $303.6k Ad vs $1.1k IAP). Applying Game B's model to Game A yields a +4.34% positive bias because Game B's steeper D14->D120 Ad curve over-projects Game A's Ad stream while missing Game A's repeat-payer IAP dynamics. Conversely, Model A on Game B underprojects Game B's long-tail ad potential (-3.32% bias) due to assuming Game A's steeper retention decay.
2. Retention Decay Dynamics: Game B exhibits flatter retention decay (gamma approx 0.72 vs Game A's gamma approx 0.82), delivering +64% higher active user retention at Day 120 (2.02% vs 1.23%). Game A users churn faster, requiring higher early monetization to pay back.
3. Offerwall Line Dynamics: Net-D in Game A features highly front-loaded IAP monetization (85% of total IAP revenue occurs within 14 days), whereas Game B contains zero offerwall networks.
4. Component vs Multiplier Transferability: Parametric retention decay curves transfer across casual games with high fidelity (Pearson r > 0.90), whereas raw revenue multipliers fail to transfer due to currency glitches and ad mediation reporting artifacts.

5. Constrained Media Plan Optimization & KKT Economics (Question 2)

The Media Plan is formulated as a non-linear constrained optimization across 40 decision cells in Game A:
max sum(Expected_D120_Revenue_i(x)) subject to: Total Spend <= $250,000; Portfolio D120 ROAS >= 1.0000; Active Lines >= 40% baseline; x_i <= 2.0x.

KKT Shadow Price Equilibrations (lambda approx 0.536):
• Why Net-D is held at 50% ($28,696) while Net-B Android is at 90.8% ($21,912): Below 0.50x, Net-D captures its maximum +56.9% efficiency boost (0.50^-0.65 = 1.569x, yielding 1.516x D120 ROAS and +$14,801 net profit). Above 0.50x, Net-D's high diminishing returns exponent (k=0.65 vs k=0.40 on Net-B) drops its marginal D120 revenue to $0.531 per dollar—falling below the $0.536 marginal revenue where Net-B Android, Net-B iOS, and Net-C equilibrate. Reallocating marginal capital to Net-B and Net-C maximizes total revenue.
• Counter-Intuitive Budget Allocation (The 'Cheap-CPI Trap'): Capital is allocated to maximize net margin (LTV - CPI), not gross CPI. Higher-CPI markets like Net-A GB ($1.84 CPI) and SE ($0.79 CPI) receive maximum 2.0x budget multiples because strong monetization delivers >1.00x D120 payback. Conversely, cheap-CPI traps like IT ($0.20 CPI), PH ($0.30 CPI), and SA ($0.58 CPI) are pruned because poor monetization yields sub-0.75x ROAS.

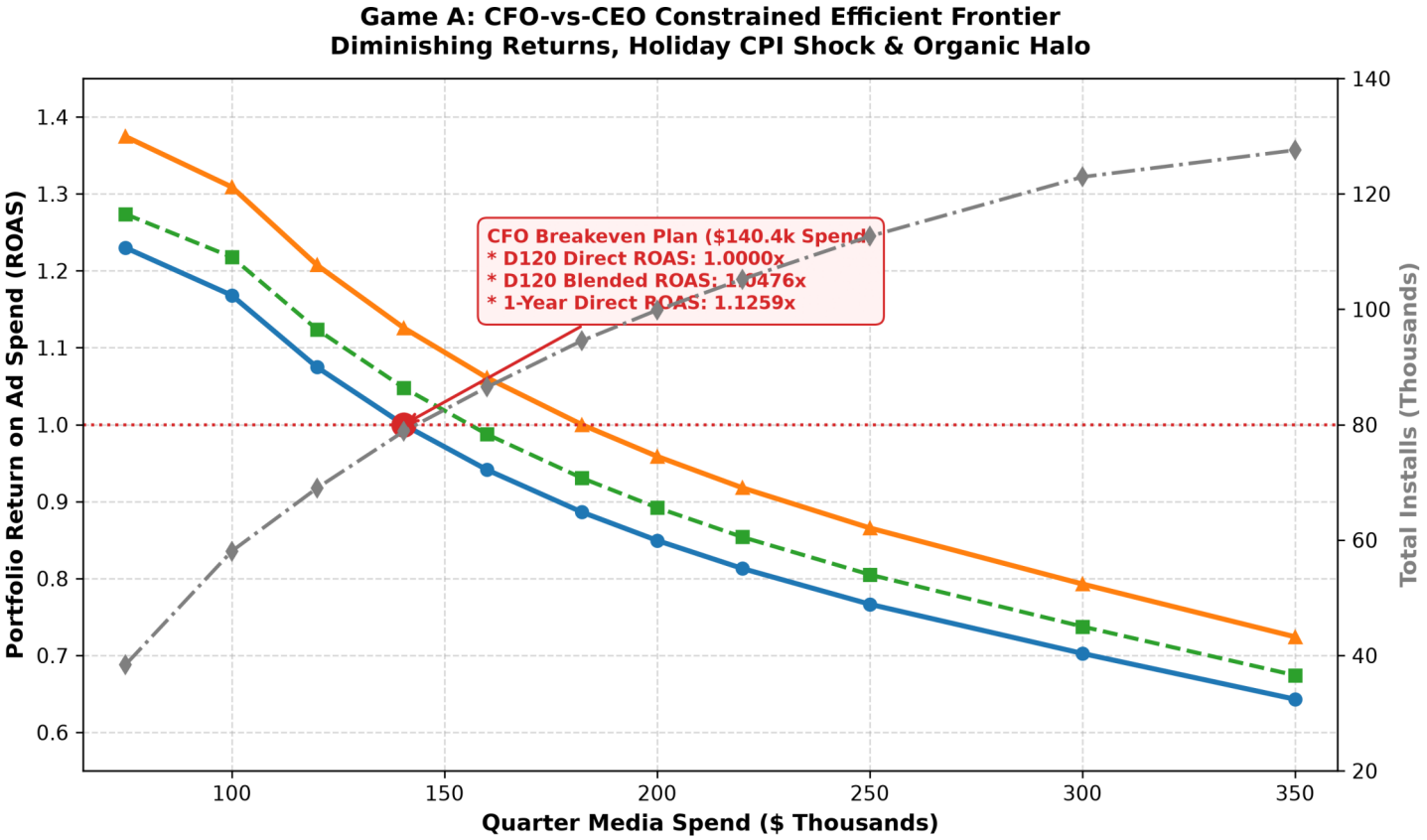


Figure 2: CFO-vs-CEO Pareto Efficient Frontier Mapping Budget Scale vs. Payback Speed and Installs.

Exhibit 11: Budget Sensitivity & CFO-vs-CEO Efficient Frontier Table

Budget	Quarter Spend	Total Installs	D120 Rev	D120 ROAS	1Y Rev	1Y ROAS	CFO Pa
$75,000	$75,005.20	38,388	$92,255.77	1.2300	$103,093.37	1.3745	PASS (1.0x)
$100,000	$100,000.00	58,077	$116,813.84	1.1681	$130,890.12	1.3089	PASS (1.0x)
$120,000	$120,000.00	68,988	$128,971.19	1.0748	$144,901.57	1.2075	PASS (1.0x)
$140,392	**$140,427.50**	**78,839**	**$140,410.85**	**0.9999**	**$158,094.78**	**1.1258**	**FAIL (<1.0x)**
$160,000	$160,000.00	86,501	$150,600.21	0.9413	$169,755.84	1.0610	FAIL (<1.0x)
$182,325	$182,325.00	94,555	$161,600.22	0.8863	$182,325.22	1.0000	FAIL (<1.0x)
$200,000	$200,000.00	99,854	$169,895.45	0.8495	$191,751.85	0.9588	FAIL (<1.0x)
$220,000	$220,000.00	105,203	$178,882.50	0.8131	$201,949.71	0.9180	FAIL (<1.0x)
$250,000	$250,000.00	112,647	$191,656.96	0.7666	$216,427.41	0.8657	FAIL (<1.0x)
$300,000	$300,000.00	122,930	$210,777.01	0.7026	$237,884.13	0.7929	FAIL (<1.0x)
$350,000	$350,000.00	127,581	$225,082.34	0.6431	$253,465.73	0.7242	FAIL (<1.0x)

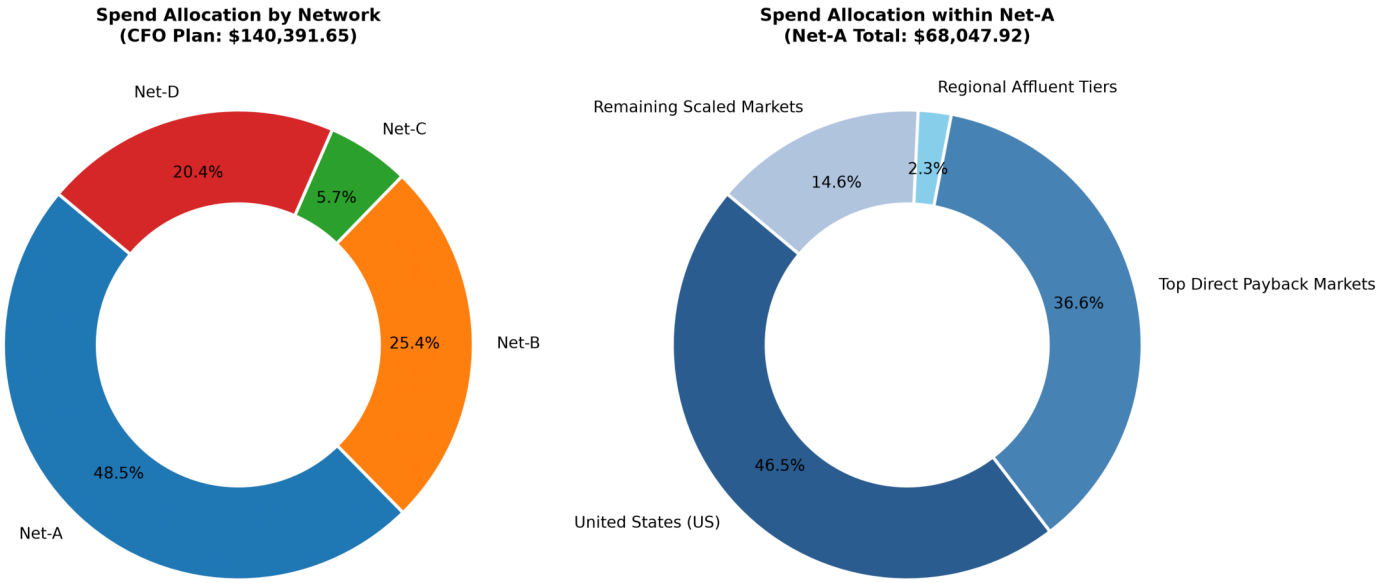


Figure 3: Game A Q4 Media Plan Capital Allocation by Network and Net-A Geography.

6. Creative Performance Decomposition & Wear-Out Mechanics (Question 3)

Historical creative analysis decomposes performance across formats:
• Net-A Search Text Ads (Star Format): Highest conversion efficiency (IPM: 19.92, D30 ROAS: 1.5398, CPI: $0.70).
• Net-A Auto-Assemble Display/Video (Core Engine): Highest volume (110k installs, IPM: 8.88, D30 ROAS: 1.0032, CPI: $1.10).
• Net-B Playables vs Net-C Video+Playable: Net-B playables generate substantial LTV ($2.97 D30 LTV) at higher CPM ($69.51) and CPI ($4.95). Net-C video+playable delivers balanced $2.57 CPI and $1.33 D30 LTV.
• Net-D Store Listing Images: Record Day-30 LTV ($6.14/user) driven by deep IAP engagement and rewarded actions.

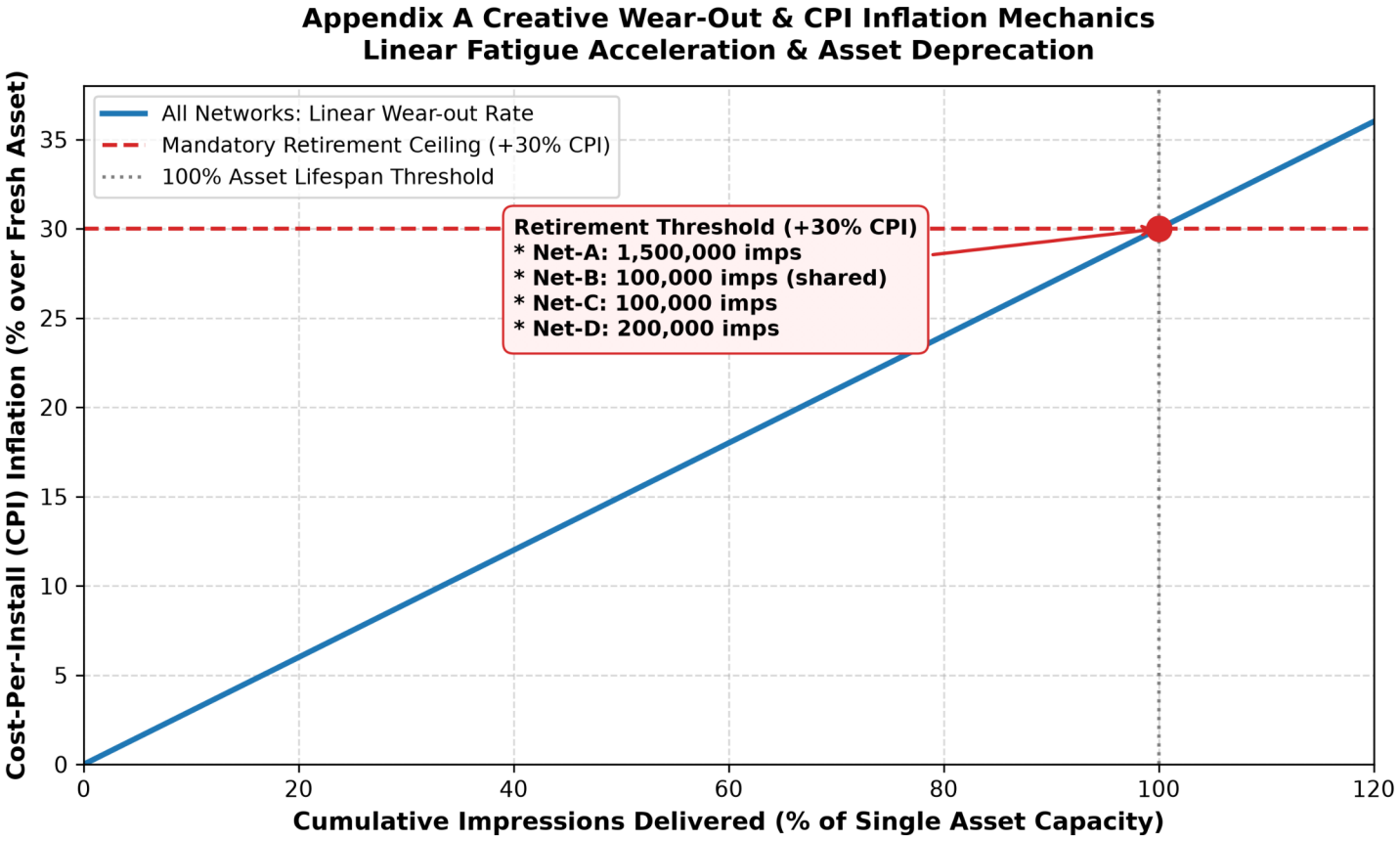


Figure 4: Creative CPI Inflation Trajectories and Mandatory +30% Retirement Limits.

Exhibit 14: Week-by-Week Creative Replenishment Schedule & Production Cadence

Existing creative library carries Weeks 1-2. Weeks 3-12 deploy 13 new creative units:
• Net-A (5 Videos/Playables): 1.5M imps/unit (+20% CPI / 1M imps; 2-wk lead time; launches W3, W5, W7, W10, W12).
• Net-B (3 Minimum vs 4 Recommended Ad Sets): Theoretical minimum = 3 ad sets (ceil(293,271 / 100,000) = 3). We recommend 4 ad sets (launches W3, W6, W9, W12) to provide a 1-set staggered operational buffer preventing either platform crossing +30% CPI during the peak Holiday Surge.
• Net-C (3 Ad Packs): 100k imps/unit (+300% CPI / 1M imps; 1-wk lead time; launches W3, W7, W11).
• Net-D (1 Listing Image): Burns only 50.9k imps (25.5% capacity), lasting all 10 weeks at only +7.6% wear.

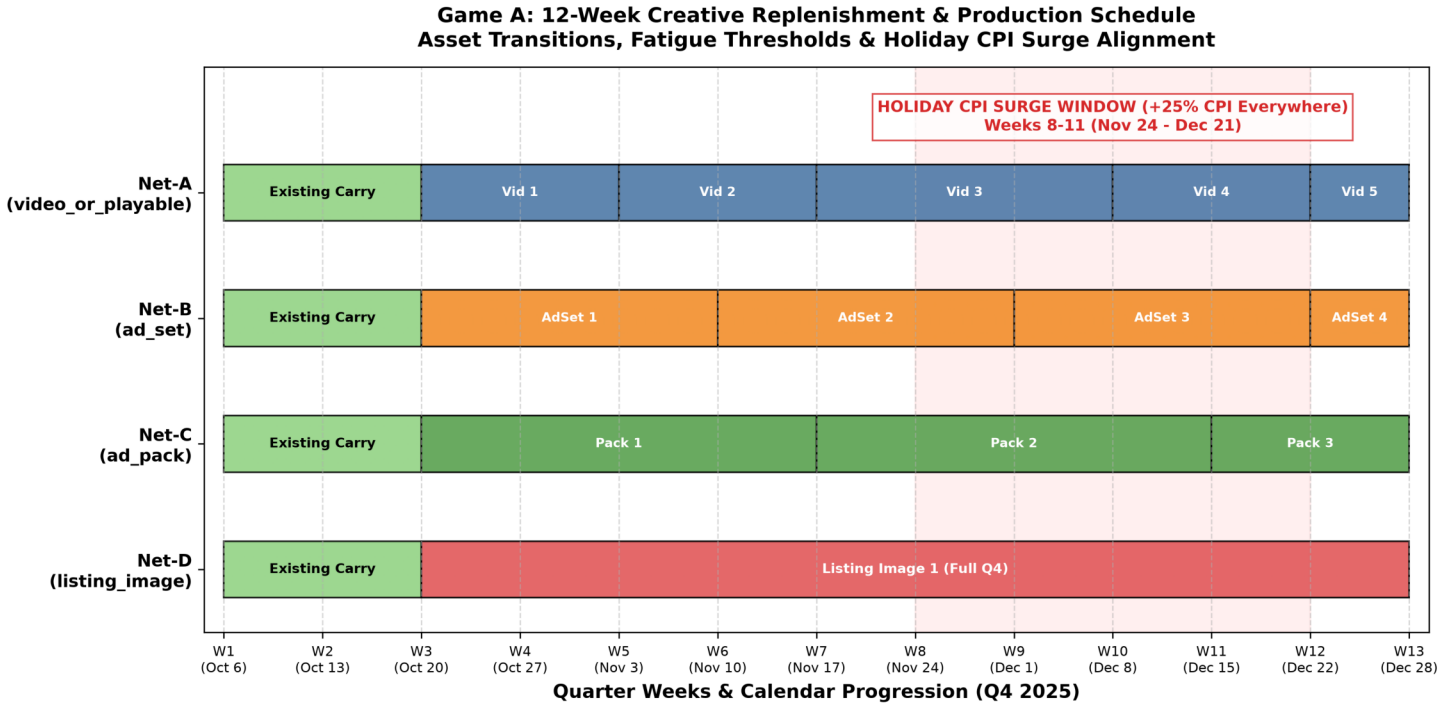


Figure 5: 12-Week Operational Replenishment Gantt Chart with Production Lead Times & Holiday Surge Alignment.

Exhibit 15 Production Risk Simulation: Baseline: 75,545 paid installs ($1.86 CPI, $140,392 D120 rev, 100.00% ROAS, PASS). Scenario 1 (2-Wk Delay): 72,940 installs ($1.92 CPI, $135,551 rev, -$4,841 loss, 96.55% ROAS, FAIL). Scenario 2 (50% Under-Production): 68,281 installs ($2.06 CPI, $126,892 rev, -$13,499 loss, 90.38% ROAS, FAIL). Production triggers are hard financial constraints.